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HPV epidemiology and *PAX1/JAM3* methylation in cervical cancer screening

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Abstract

Persistent high-risk human papillomavirus (hrHPV) infection drives cervical carcinogenesis, yet accurate biomarkers of high grade cervical intraepithelial neoplasia (CIN2/3) are needed to optimize HPV-based cervical screening. The prevalence of hrHPV types specifically in Changshu was assessed, and the roles of *PAX1/JAM3* gene methylation ($PAX1^m/JAM3^m$) in finding cervical lesions were studied. 52,605 women aged 30–65 years were screened by HPV genotype detection. Furthermore, HPV-positive women were stratified by age and region for subgroup analysis; among them, 173 underwent histopathological assessment, quantitative viral load measurement, and methylation-specific qPCR for *PAX1* and *JAM3*. The diagnostic performance of HPV 16/18 genotyping, $PAX1^m$ and $JAM3^m$ was compared. We also performed logistic regression models to determine independent predictors of CIN2/3. The overall prevalence of HPV was 6.17% (3245/52605); HPV52 (22.69%, 907/3997), HPV58 (11.56%, 462/3997), and HPV16 (10.13%, 405/3997) were the three most prevalent genotypes. Single genotype infection was detected in 2636 cases (2636/3245, 81.23%), while multiple HPV infections were observed in 609 cases (18.77%). Among multiple HPV infections, 489 cases had two genotypes, 100 had three genotypes, 17 had four genotypes, and 3 had five genotypes. HPV infection prevalence increased with age ($\chi^2 = 101.3$, $P<0.0001$), peaking at age over 60 years, and differed among sub-districts ($\chi^2 = 61.53$, $P<0.0001$). Among the 173 follow-up women, including 44 normal lesions, 101 CIN1 lesions, 15 CIN2 lesions, and 13 CIN3 lesions, $PAX1^m$ and $JAM3^m$ levels increased stepwise ($P<0.0001$). Moreover, $PAX1^m/JAM3^m$ had a robust positive association ($R=0.519$, $P<0.0001$), but both were independent of viral load. For CIN3, the dual methylation achieved high diagnostic potential, with an area under the curve (AUC) of 0.92. For age-adjusted multivariable analysis, the weighted composite methylation score derived from *PAX1* and *JAM3* was an independent risk factor for CIN2+lesions (adjusted OR=1.965, 95% CI: 1.520–2.541, $P<0.001$). The study delineates the HPV prevalence characteristics in the Changshu population. Moreover, it demonstrates that $PAX1^m/JAM3^m$ are viral load-independent biomarkers of CIN2/3 lesions and significantly enhance the diagnostic performance of primary HPV screening.

Keywords HPV · *PAX1* · *JAM3* · Methylation · Cervical cancer

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Introduction

Cervical cancer (CC) continues to be a major public health issue in the world's women, with incidence increasing steadily in the past decade. CC is the fourth leading malignancy in women worldwide, constituting 6.8% of new cancer diagnoses and 8.1% of annual cancer-attributable deaths[1]. CC is largely preventable and curable when detected and treated at its precancerous stages[2]. Therefore, timely diagnosis and intervention of cervical precancerous lesions are important for CC prevention. Persistent hrHPV infection is responsible for the majority of CC and high-grade squamous intraepithelial lesions (HSILs)[3]. However, the clinical utility of HPV viral load for triaging HPV-positive women during primary cervical screening remains controversial. A retrospective study in Beijing observed a significant elevation of HPV16 viral load alongside progressive cervical lesion severity using TCT and colposcopic assessments; HPV33 demonstrated a strong association with severe lesions in TCT screening, whereas HPV52 and HPV58 were more frequently linked to low-grade abnormalities; Notably, HPV18 viral load showed no statistically significant correlation with disease severity[4]. Despite the above findings, the current HPV screening strategy has some limitations. HPV DNA testing does not distinguish transient infection from those infections that may progress—providing a clinical dilemma of unnecessary colposcopies and overtreatment for screen positive women. Therefore, the World Health Organization's 2020 global strategy to accelerate the CC elimination set out 2030 targets based on a tripartite prevention strategy[5]. Effective population level screening is critical forsecondary prevention. Accordingly, there is an urgent need to identify screening biomarkers that combine precision, ease of implementation, and improved diagnostic accuracy.

Recently, there has been growing interest in DNA methylation as an epigenetic biomarker to facilitate early CC detection and risk stratification. Paired box gene 1 (PAX1), important for embryonic development, normally has a low degree of methylation in healthy cervical tissue[6,7]. Yet, during carcinogenesis from intraepithelial neoplasia to invasive cancer, its promoter acquires aberrant hypermethylation, gene expression is silenced, and proliferation constraints are removed[8,9]. Junctional adhesion molecule 3 (JAM3), which is important for maintenance of intercellular tight junctions, is similarly highly methylated with a correlation to increased tumor invasiveness and metastasis[10,11]. Combined *PAX1* and *JAM3* methylation was markedly elevated in CC tissues than in benign tissues, and this methylation level increased with histopathological severity[12,13], positioning the dual marker panel as a promising biological tool for early screening and risk assessment.

In conclusion, understanding how the $PAX1^m$ and $JAM3^m$ expression patterns of CC cells relate to viral burden will deepen our knowledge of CC carcinogenesis and may enable more refined screening triage. Therefore, in the present study, we profiled HPV prevalence among the Changshu population, assessed the methylation level of the two genes, and explored their correlations with HPV viral load, laying a foundation for personalized early detection.

Methods

Study subjects and sample collection

Women aged 30–65 years were recruited for specimen collection. A cervical brush was rotated clockwise 3–6 times against the cervical mucosa to collect enough exfoliated cells. The brush was carefully retrieved and vortexed within collection tubes to dislodge cells. HPV genotyping was performed timely. The study received ethical approvalfrom the Biomedical Ethics Committee of Changshu Medical Laboratory (JYS-LL-2024-011). Informed consent was obtained from participants. All procedures followed applicable guidelines and regulations.

HPV genotyping

DNA was extracted from cervical cells using pre-dispensed plates. We amplified 14 hrHPV genotypes (HPV16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, 66, 68) using SLAN-96 S PCR System (Hongshi Medical, Shanghai, China).

DNA methylation analysis

Bis-DNA was bound to magnetic beads (600 µL binding buffer + 50 µL bead suspension), washed twice (400 µL wash buffer, 3 min on a magnet), and treated with 200 µL processing buffer (20–30 ℃ for 15 min with gentle tube tapping). After two additional washes, the beads were dried at 55 ℃ for 20 min, and DNA was subsequently eluted using 45 µl elution buffer at 55 ℃ for 4 min. The eluate was transferred to a clean tube. The methylation detection kit (Origin Biotechnology Co., Ltd., Beijing, China) was used. Each 25 µL PCR reaction contained 15 µL *PAX1/JAM3* methylation reaction mixture, 5 µL primer probe mixture, as well as 5 µL Bis-DNA template; PCR tubes were centrifuged (2000 rpm, 10 s) before thermal cycling. PCR conditions consisted of a denaturation at 96 ℃ for 10 min, 45 cycles of 94 ℃ for 15 s, 64 ℃ for 5 s, and 60 ℃ for 30 s, with a cooling at 25 ℃ for 1 min. Methylation levels were assessed by calculating ΔCt values for *PAX1* and *JAM3* as $Ct_{PAX1} - Ct_{GAPDH}$ and $Ct_{JAM3} - Ct_{GAPDH}$, respectively.

Statistical analysis

GraphPad Prism 10.0, SPSS 22.0 (IBM Corp., Armonk, NY, USA) and R (version 4.2.1) were used to complete all statistical analysis. The Shapiro-Wilk normality test and Levene's test for homogeneity of variances were carried out on the acquired data. If both normal distribution and homogeneity of variance were met, one-way ANOVA was employed; otherwise, nonparametric tests were used, and the Kruskal-Wallis H-test was adopted for multigroup comparisons. The chi-square (χ^2) test was conducted to evaluate differences in infection rates among different age groups and regions. Chord diagrams and heat maps were generated based on HPV genotype co-infection data. Two-sided statistical tests were applied, and $P < 0.05$ was defined as statistically significant.

Receiver operating characteristic curves (ROC) were used to assess the diagnostic performance of individual and combined detection of HPV16/18 genotypes, $PAX1^m$, and $JAM3^m$ for CIN2/3 lesions. The AUC and 95% confidence interval (95% CI) were calculated, and sensitivity and specificity were compared among different testing approaches. Pearson correlation analysis was applied for normally distributed variables to assess relationships of *PAX1/JAM3* methylation levels with cervical lesion pathological grades and HPV viral load; otherwise, Spearman rank correlation analysis was adopted.

To evaluate the independent contribution of the methylation markers to cervical intraepithelial neoplasia grade 2 or worse (CIN2+), we performed binary logistic regression analysis. Given the limited number of positive outcomes ($n = 28$) and to avoid overfitting, we constructed a composite methylation score using weighted summation of the ΔCt values of *PAX1* and *JAM3*. The weights were derived from the β-coefficients of univariate logistic regression models for each marker ($\beta_PAX1 = -0.222$; $\beta_JAM3 = -0.442$). The formula was defined as methylation score = $(-0.222) \times \Delta Ct_{PAX1} + (-0.442) \times \Delta Ct_{JAM3}$.

Results

HPV genotyping result

Between March and August 2024, a total of 52,605 cervical samples were subjected to HPV genotyping, among which 3245 (6.17%) *were* HPV positive. A subset of 173 specimens was enrolled for further follow-up analyses,

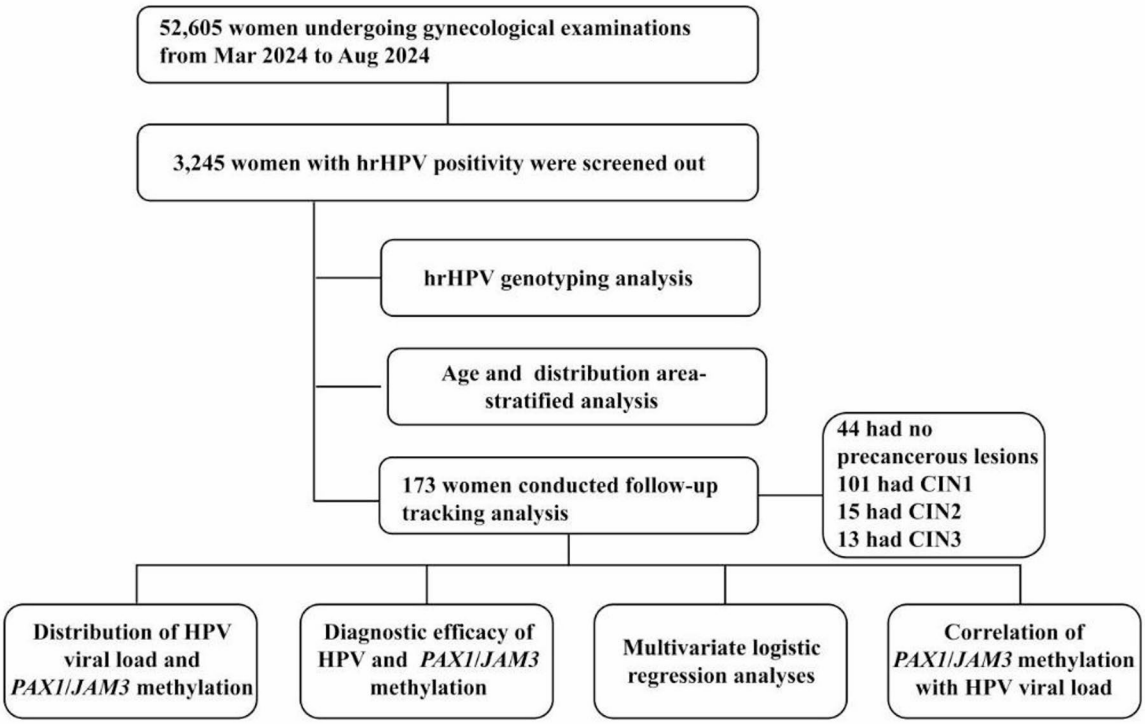


Fig. 1 The overall research workflow diagram.

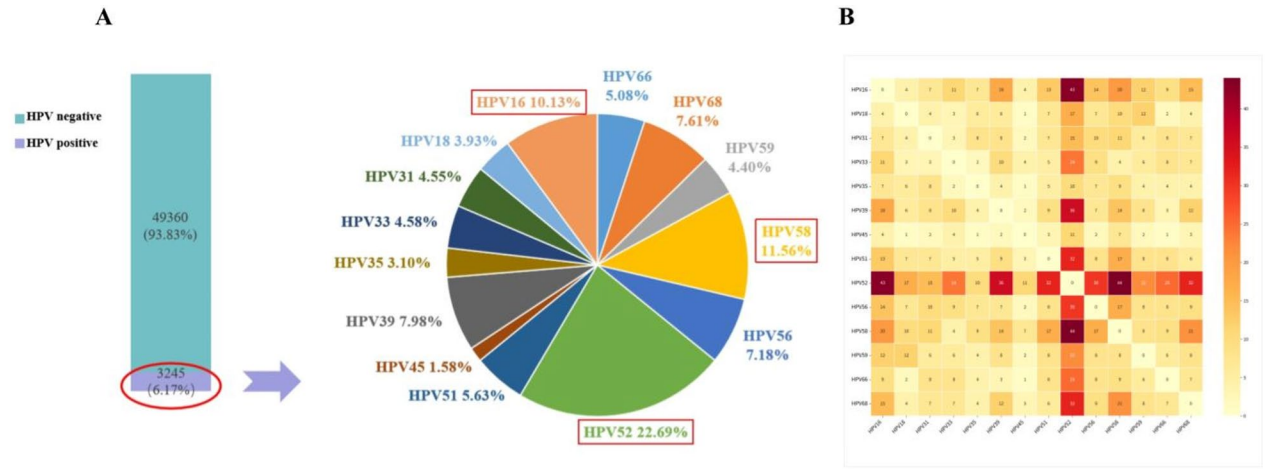


Fig. 2 HPV genotype distribution and co-infection correlations among women with multiple HPV infections. (**A**) Overall distribution of HPV genotypes among all 52,605 screening women. (**B**) Co-infection correlations among multiple HPV infections.

including histopathological examination, quantitative HPV viral load measurement, and *PAX1*/*JAM3* gene methylation detection (study workflow, Fig. 1).

Among a total of 3997 detected HPV genotypes from 3245 HPV-positive cases, HPV52 was the most prevalent (22.69%), followed by HPV58 (11.56%), HPV16 (10.13%), HPV39 (7.98%), HPV68 (7.61%), HPV56 (7.18%), HPV51 (5.63%), HPV66 (5.08%), HPV33 (4.58%), HPV31 (4.55%), HPV59 (4.40%), HPV18 (3.93%), HPV35 (3.10%), and HPV45 (1.58%) (Fig. 2A). 2636 (81.23%) cases harbored only a single genotype. The five most prevalent genotypes were HPV52 ($n=651$), HPV58 ($n=315$), HPV16 ($n=271$), HPV39 ($n=216$), and HPV68

($n=199$). The remaining 609 (18.77%) cases presented multiple infections: 489 with two genotypes (the most prevalent pair was 52 and 58, $n=26$), 100 with three genotypes, 17 with four genotypes, and 3 with five genotypes. Among all multiple genotype infections, the most frequent genotype pairs were HPV52 + HPV58, followed by HPV16 + HPV52 and HPV39 + HPV52. The strength of association among individual HPV types within multiple-type infections is shown in Fig. 2B.

Age-stratified analysis of HPV prevalence

Participants were divided into seven age groups. The overall HPV prevalence increased with age, peaking in the over 60-years age group ($\chi^2 = 101.3$, $P<0.0001$; Table 1; Fig. 3A). For single genotype of HPV infection, the prevalence trend was largely paralleled the overall prevalence (Fig. 3B). For multiple HPV genotype infections, double infections were lowest in 46–50 year, while the highest level in the over 60 year (Fig. 3C). Triple infections exhibited a U-shaped distribution (Fig. 3D), and quadruple and quintuple infections were low in younger age groups and increased modestly after 56 years of age (Fig. 3E-F).

Geographic distribution of HPV subtypes

Our analysis showed a substantial geographic variation in the overall HPV prevalence stratified by sub-district (4.75%–8.62%), with the highest prevalence in the Southeast sub-district and the lowest prevalence in the Yushan sub-district ($\chi^2 = 61.53$, $P<0.0001$; Table 2). HPV52 was the predominant genotype across all sub-districts, accounting for 22.69% (907/3997) of all positive cases, followed by HPV58 (11.56%, 462/3997) and HPV16 (10.13%, 405/3997).

Notably, HPV52 exhibited the highest prevalence in Southeast Sub-district (25.64%) and Changfu sub-district (24.70%), whereas presenting relatively low prevalence in the Bixi (18.73%) and Mocheng sub-district (19.73%). The prevalence of HPV 16 showed great regional differences, from 5.82% in the Yushan sub-district to 13.07% in the Bixi sub-district. Interestingly, despite its relatively small sample size, the Southeast sub-district displayed higher proportions of HPV58 (15.38%), HPV39 (10.26%), and HPV51 (10.26%) compared with other regions. HPV 18 was especially common in the Yushan sub-district (6.55%) than the other areas (0.00% to 4.62%).

Table 1 Distribution of HPV-positive genotypes across different age groups.

HPV genotype	<36 (%)	36–40 (%)	41–45 (%)	46–50 (%)	51–55 (%)	56–60 (%)	>60 (%)	Total
HPV 52 (%)	20 (19.05%)	77 (22.92%)	116 (25.84%)	113 (23.44%)	180 (22.19%)	261 (22.44%)	140 (21.51%)	907
HPV 58 (%)	12 (11.43%)	37 (11.01%)	60 (13.36%)	60 (12.45%)	96 (11.84%)	131 (11.26%)	66 (10.14%)	462
HPV 16 (%)	12 (11.43%)	36 (10.71%)	43 (9.58%)	48 (9.96%)	78 (9.62%)	117 (10.06%)	71 (10.91%)	405
HPV 39 (%)	12 (11.43%)	30 (8.93%)	23 (5.12%)	42 (8.71%)	71 (8.75%)	91 (7.82%)	50 (7.68%)	319
HPV 68 (%)	4 (3.81%)	22 (6.55%)	33 (7.35%)	38 (7.88%)	57 (7.03%)	97 (8.34%)	53 (8.14%)	304
HPV 56 (%)	12 (11.43%)	27 (8.04%)	31 (6.90%)	17 (3.53%)	65 (8.01%)	85 (7.31%)	50 (7.68%)	287
HPV 51 (%)	5 (4.76%)	20 (5.95%)	32 (7.13%)	18 (3.73%)	49 (6.04%)	68 (5.85%)	33 (5.07%)	225
HPV 66 (%)	3 (2.86%)	20 (5.95%)	21 (4.68%)	33 (6.85%)	34 (4.19%)	59 (5.07%)	33 (5.07%)	203
HPV 33 (%)	4 (3.81%)	19 (5.65%)	14 (3.12%)	20 (4.15%)	24 (2.96%)	61 (5.25%)	41 (6.30%)	183
HPV 31 (%)	5 (4.76%)	12 (3.57%)	27 (6.01%)	23 (4.77%)	39 (4.81%)	51 (4.39%)	25 (3.84%)	182
HPV 59 (%)	8 (7.62%)	12 (3.57%)	15 (3.34%)	19 (3.94%)	39 (4.81%)	51 (4.39%)	32 (4.92%)	176
HPV 18 (%)	3 (2.86%)	13 (3.87%)	9 (2.00%)	28 (5.81%)	34 (4.19%)	44 (3.78%)	26 (3.99%)	157
HPV 35 (%)	3 (2.86%)	8 (2.38%)	13 (2.90%)	17 (3.53%)	28 (3.45%)	34 (2.92%)	21 (3.23%)	124
HPV 45 (%)	2 (1.90%)	3 (0.89%)	12 (2.67%)	6 (1.24%)	17 (2.10%)	13 (1.12%)	10 (1.54%)	63
HPV positive	105	336	449	482	811	1163	651	3997
NO. of positive cases	84 (4.59%)	283 (5.84%)	386 (5.20%)	428 (5.36%)	667 (5.72%)	903 (6.92%)	494 (8.50%)	3245
Total cases	1831	4846	7426	7980	11,662	13,050	5810	52,605

Note: "HPV positive" refers to cumulative count of detected HPV genotypes; "No. of positive cases" refers to the number of infected participants. In short, the former value exceeds, as a single participant may be infected with multiple HPV genotype infections.

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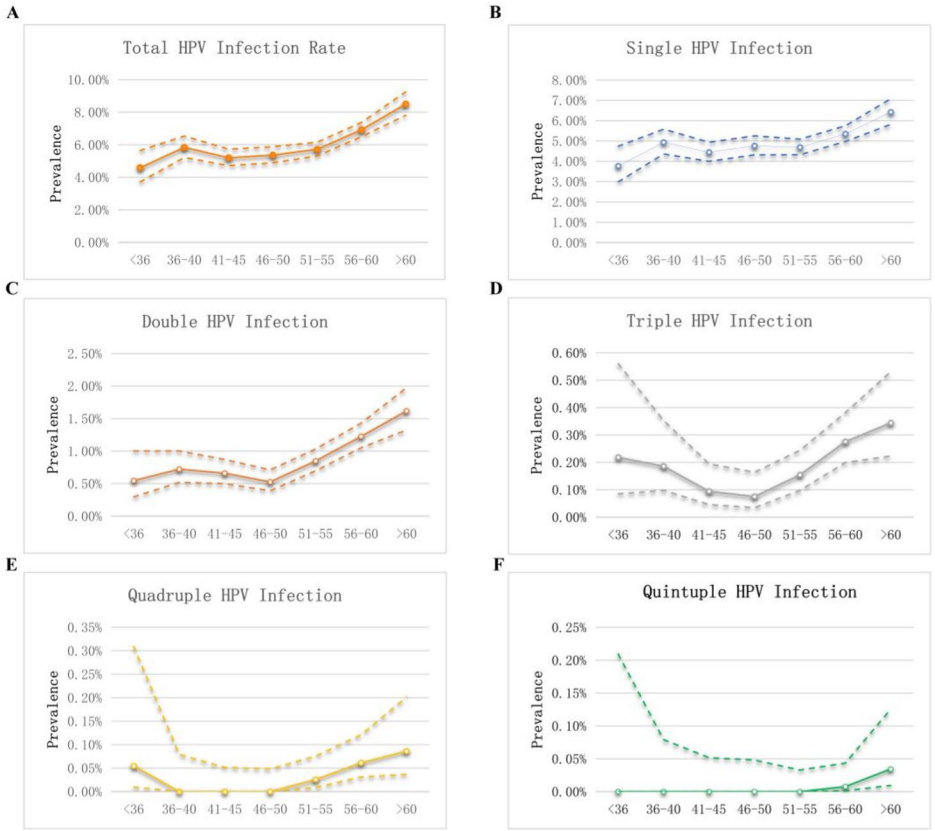


Fig. 3 Age specific prevalence of HPV infection among women. (**A**) Overall HPV infection rate across different age groups. (**B**) Single-type HPV infection rate stratified by age groups. (**C**) Double-type HPV infection rate stratified by age groups. (**D**) Triple-type HPV infection rate stratified by age groups. (**E**) Quadruple-type HPV infection rate stratified by age groups. (**F**) Quintuple-type HPV infection rate stratified by age groups. The dashed line refers to the 95% CIs.

Table 2 Positive distribution of HPV genotyping among populations in different areas.

HPV genotype	Bixi sub-district (%)	Changfu sub-district (%)	Southeast sub-district (%)	Mocheng sub-district (%)	Yushan sub-district (%)	Qinchuan sub-district (%)	Towns (%)	Others (%)	Total
HPV 52 (%)	53(18.73%)	83(24.70%)	10(25.64%)	29(19.73%)	58(21.09%)	85(22.73%)	477(24.09%)	112(19.89%)	907
HPV 58 (%)	33(11.66%)	31(9.23%)	6(15.38%)	16(10.88%)	41(14.91%)	39(10.43%)	231(11.67%)	65(11.55%)	462
HPV 16 (%)	37(13.07%)	31(9.23%)	3(7.69%)	16(10.88%)	16(5.82%)	46(12.30%)	197(9.95%)	59(10.48%)	405
HPV 39 (%)	24(8.48%)	21(6.25%)	4(10.26%)	16(10.88%)	21(7.64%)	41(10.96%)	153(7.73%)	44(7.82%)	319
HPV 68 (%)	18(6.36%)	22(6.55%)	2(5.13%)	12(8.16%)	14(5.09%)	21(5.61%)	164(8.28%)	47(8.35%)	304
HPV 56 (%)	25(8.83%)	32(9.52%)	2(5.13%)	11(7.48%)	26(9.45%)	18(4.81%)	130(6.57%)	46(8.17%)	287
HPV 51 (%)	13(4.59%)	14(4.17%)	4(10.26%)	10(6.80%)	14(5.09%)	26(6.95%)	97(4.90%)	45(7.99%)	225
HPV 66 (%)	12(4.24%)	16(4.76%)	0(0.00%)	9(6.12%)	15(5.45%)	24(6.42%)	102(5.15%)	27(4.80%)	203
HPV 33 (%)	22(7.77%)	16(4.76%)	3(7.69%)	8(5.44%)	13(4.73%)	13(3.48%)	95(4.80%)	19(3.37%)	183
HPV 31 (%)	9(3.18%)	26(7.74%)	2(5.13%)	8(5.44%)	18(6.55%)	16(4.28%)	87(4.39%)	16(2.84%)	182
HPV 59 (%)	13(4.59%)	14(4.17%)	1(2.56%)	7(4.76%)	12(4.36%)	15(4.01%)	85(4.29%)	27(4.80%)	176
HPV 18 (%)	6(2.12%)	12(3.57%)	0(0.00%)	2(1.36%)	18(6.55%)	14(3.74%)	71(3.59%)	26(4.62%)	157
HPV 35 (%)	12(4.24%)	7(2.08%)	1(2.56%)	2(1.36%)	7(2.55%)	13(3.48%)	60(3.03%)	22(3.91%)	124
HPV 45 (%)	6(2.12%)	11(3.27%)	1(2.56%)	1(0.68%)	2(0.73%)	3(0.80%)	31(1.57%)	8(1.42%)	63
HPV positive	283	336	39	147	275	374	1980	563	3997
NO. of positive cases	237(5.88%)	267(6.66%)	35(8.62%)	118(5.48%)	232(4.75%)	306(5.34%)	1585(6.92%)	465(5.48%)	3245
Total	4028	4011	406	2154	4889	5728	22,899	8490	52,605

Notes: "Others" means that other people in the tested region have no registered screening information. "HPV positive" indicates the total number of HPV positives of different types; "No. of positive cases" is the number of positive cases. Consequently, the former is greater than the latter, as one individual could have more than one HPV type.

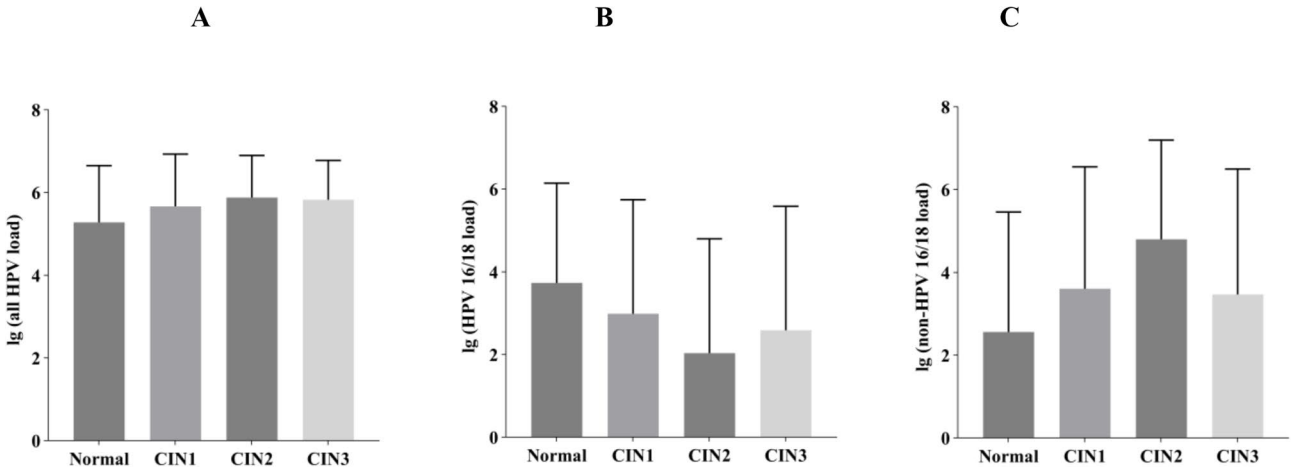


Fig. 4 Column chart of HPV viral load across different cervical lesions. (**A**) Total HPV viral load stratified by cervical histopathological grade. (**B**) HPV 16/18 viral load across different histopathological grades. (**C**) Non-16/18 HPV viral load distribution based on histopathological status.

Table 3 Detection of *PAX1*^{m} and *JAM3*^{m} across different grades of cervical lesions.

Group	Total	Age	ΔCt PAX1	ΔCt JAM3
Normal	44	54.50 (46.50, 59.00)	14.23 (8.13, 18.92)	9.93 (9.17, 10.99)
CIN1	101	55.00 (47.00, 59.50)	12.63 (9.19, 18.99)	10.04 (9.20, 10.87)
CIN2	15	55.00 (46.00, 60.00)	7.39 (5.15, 16.56)	9.28 (6.02, 10.05)
CIN3	13	45.00 (37.50, 58.00)	6.85 (3.41, 7.33)	7.37 (4.59, 9.28)
H		3.71	26.56	23.37
P		0.29	**<0.0001**	**<0.0001**

Association of histopathology with HPV viral load and gene methylation

Among the 173 women follow-up women, histology evaluation identified 44 normal cases, 101 CIN1 lesions, 15 CIN2 lesions, and 13 CIN3 lesions. HPV viral load analysis across different cervical histopathological grades revealed no significant differences in all HPV, HPV16/18, or non-HPV16/18 infections (Fig. 4A-C). Both *PAX1*^{m} and *JAM3*^{m} exhibited a significant positive gradient across histological groups. The higher methylation levels were observed in CIN3 lesions than in normal/CIN1 tissues (both $P<0.0001$; Table 3), indicating that methylation levels were likely positively correlated with severity of disease.

Diagnostic accuracy evaluation

For CIN1 and CIN2 lesions, the dual methylation exhibited overall diagnostic accuracy comparable to HPV16/18 in terms of AUC. It achieved higher sensitivity but lower specificity for CIN1 lesions, and higher specificity but lower sensitivity for CIN2 lesions. When detecting CIN3 lesions, HPV16/18 genotyping exhibited a sensitivity of 54% and a specificity of 75%. Both individual methylation markers outperformed HPV16/18 for CIN3 detection: *PAX1* achieved a sensitivity/specificity of 92%/80%, whereas *JAM3* achieved 69%/96%. The dual marker further improved diagnostic performance, with a sensitivity of 92% and a specificity of 82% (Table 4). Collectively, these findings indicated that the *PAX1/JAM3* dual methylation provided better detection efficacy for CIN2 + lesions, especially CIN3, compared with HPV 16/18 genotyping.

Table 4 Diagnostic efficacy of HPV 16/18 and *PAX1*^{m} and *JAM3*^{m} detection in the screening of cervical lesions.

Indicators	CIN1			CIN2			CIN3		
	AUC (95% CI)	SE	SP	AUC (95% CI)	SE	SP	AUC (95% CI)	SE	SP
HPV 16/18	0.59 (0.49, 0.69)	0.43	0.75	0.68 (0.51, 0.84)	0.60	0.75	0.64 (0.47, 0.82)	0.54	0.75
ΔCt *PAX1*	0.53 (0.43, 0.63)	0.79	0.34	0.67 (0.49, 0.86)	0.53	0.84	0.88 (0.79, 0.98)	0.92	0.80
ΔCt *JAM3*	0.51 (0.41, 0.62)	0.65	0.46	0.70 (0.55, 0.86)	1.00	0.32	0.86 (0.73, 0.98)	0.69	0.96
ΔCt *PAX1* + ΔCt *JAM3*	0.54 (0.44, 0.64)	0.71	0.41	0.71 (0.56, 0.86)	0.33	0.98	0.92 (0.85, 0.99)	0.92	0.82

Table 5 Multivariable logistic regression analysis for indicators associated with CIN2+.

Indicators	B	S.E.	Wald	*P*	Adjusted OR	95% CI
Age	-0.069	0.030	5.267	0.022	0.933	0.880–0.990
Methylation score	0.676	0.131	26.575	<0.001	1.965	1.520–2.541

Multivariate logistic regression analysis

To determine independent variables of CIN2 + lesions, a multivariable logistic regression model was constructed, including weighted composite methylation score and age as continuous covariates. As shown in Table 5, the methylation score remained significantly and independently associated with CIN2 + lesions (*adjusted OR* = 1.965, *95% CI*: 1.520–2.541, $P < 0.001$), indicating that a higher methylation score corresponds to an increased risk of high-grade lesions. Age was also retained in the model and showed a modest inverse association (*adjusted OR* = 0.933, *95% CI*: 0.880–0.990, $P = 0.022$). Hosmer-Lemeshow goodness-of-fit testing revealed that the model was well calibrated ($\chi^2 = 8.947$, $P = 0.347$), suggesting that the predicted probabilities were not significantly different from the observed outcomes.

Association between PAX1/JAM3 methylation levels and HPV viral load

No significant correlations were found between the *PAX1* and *JAM3* gene methylation levels and all HPV viral load, HPV16/18 viral load, or non-16/18 viral load. However, a positive correlation was identified between *PAX1* and *JAM3* methylation levels ($R = 0.526$, $P < 0.0001$, Fig. 5), suggesting that the methylation of these two genes might increase concomitantly during the progression of cervical lesions.

Discussion

To our knowledge, this is a large scale population-based HPV epidemiological study among 52,605 participants aged ≥ 30 years in Changshu, China. It profiles HPV genotype distribution across the district, with an overall prevalence rate of 6.17%. The three most predominant genotypes were HPV52, 58, and 16, which was consistent with the previous study in Putian[14]. Single genotype infections dominated (> 80% of infections), with HPV52, 58, and 16 being the most prevalent types. Among multiple infections, the HPV52/58 combination was the most common pair, followed by HPV16/52 and HPV39/52 co-infections. Stratified analysis revealed significant age disparities in HPV prevalence, peaking in the > 60 years age group. There was geographic heterogeneity in HPV prevalence observed across sub-districts, with the highest infection rates seen in the Southeast sub-district; but this elevated prevalence was likely attributed to the smaller sample size, which indicates the importance of sufficient sample sizes for establishing reliable prevalence.

HPV prevalence varies across geographic regions worldwide, ranging from 6.7% to 21.66% in different areas of China[15–18]. In the present study, the infection rate was relatively low, which might be related to many factors such as living habits, popularization of HPV screening, and the degree of vaccination coverage in different areas.

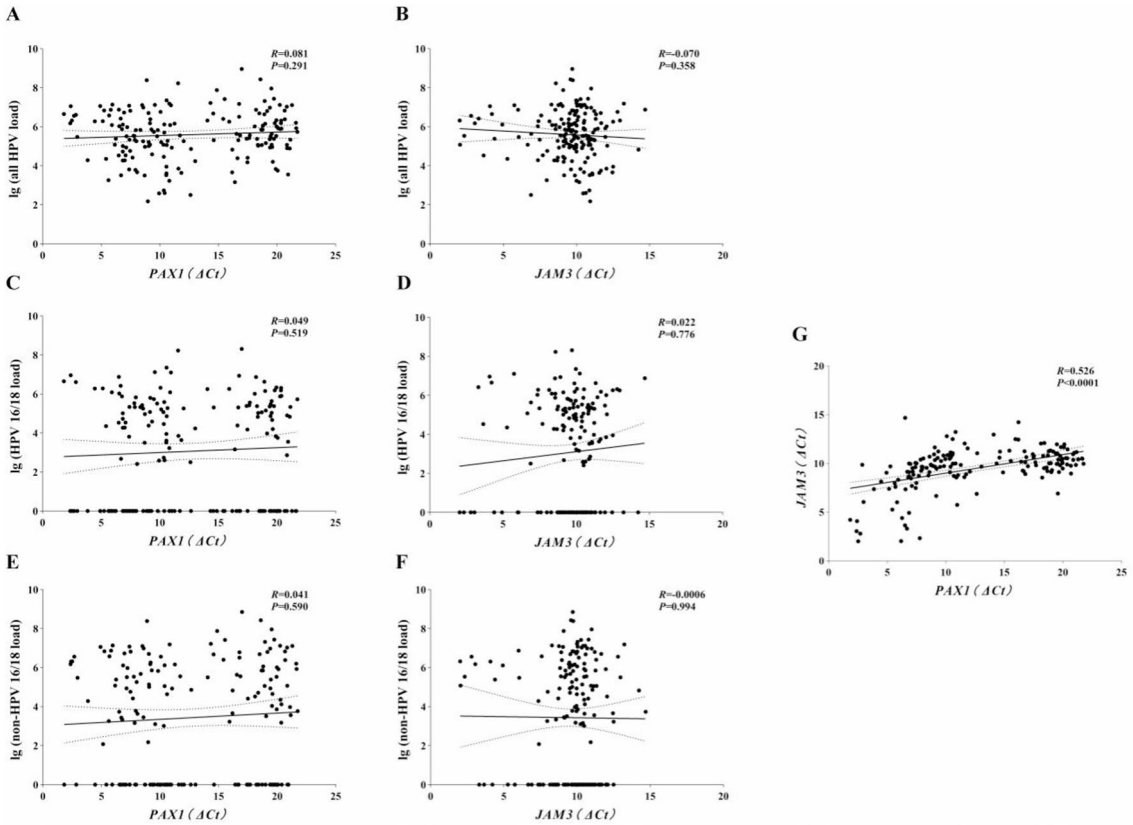


Fig. 5 Correlation analysis between $PAX1^{m}/JAM3^{m}$ and HPV viral load. (**A**) Correlation between *PAX1* methylation levels and all HPV viral load; (**B**) Correlation between *JAM3* methylation levels and all HPV viral load; (**C**) Correlation between *PAX1* methylation levels and HPV16/18 viral load; (**D**) Correlation between *JAM3* methylation levels and HPV16/18 viral load; (**E**) Correlation between *PAX1* methylation levels and non-HPV16/18 viral load; (**F**) Correlation between *JAM3* methylation levels and non-HPV16/18 viral load; (**G**) Correlation between *PAX1* and *JAM3* methylation levels.

Notably, substantial heterogeneity in HPV prevalence was observed across sub-districts of Changshu, perhaps attributable to different demographic characteristics in the local population.

This study identified age-related disparities in HPV prevalence, with infection rates rising with age and peaking in the > 60 years age group. Previous studies also reported that the HPV infection rates peak at < 25 years group, followed by a gradual decline with a smaller secondary peak at 55–59 years of age[19]. Another study reported two infection peaks at < 20 years and 51–60 years in Suzhou[20]. A study involving 38,000 subjects in Taizhou demonstrated a typical bimodal age distribution of HPV prevalence. The primary peak of 41.6% was detected among subjects younger than 21 years, and the second peak of 28.5% in the 56–60 years age group. Both peaks were markedly higher than the HPV prevalence rates among middle aged individuals[21]. As our study population did not include people < 30 years old, we did not observe the infection peak around 20 years of age. However, our study observed elevated HPV prevalence at approximately 60 years old consistent with the secondary peak observed in the above two studies. It could also be explained by diminishing immune function and vaginal micro-ecology changes of the elderly after menopause, which reduced the efficiency of their immune responses towards clearing either acquired or pre-existing HPV infections[22].

With regard to mixed HPV infections, dual infections were the most prevalent type, with HPV52/58, HPV16/52, and HPV39/52 identified as the most common pairs. These results were in agreement with the findings of a large trial in Nanjing, which reported that the most commonly identified types in outpatient women were HPV16/58, HPV52/58, and HPV39/52; in healthy women, the most common pairs were HPV16/52, HPV52/58, and HPV39/58[23]. These common mixed HPV infection genotypes were also highly similar between the two regions

and consisted of mixed infection combinations of HPV16, 39, 52, and 58. These results indicated that regional HPV prevention and control strategies should be tailored to prioritize the locally dominant HPV genotypes.

For 173 women follow-up women, further determinations of methylation levels and HPV viral load were conducted. Our study evaluated the diagnostic performance for HPV viral load and $PAX1^m$ and $JAM3^m$ for different cervical squamous epithelium lesions, and aimed to provide a more optimized triage testing protocol. Our results did not show obvious differences in HPV viral load between cervical histological grades, which was inconsistent with previous studies[24,25]. Due to the cross-sectional design, we could not assess the HPV infection duration. Persistent HPV infection is a critical determinant of cervical lesion progression. Therefore, HPV viral load without confirmed infection duration can't distinguish the different grades of lesions effectively. Moreover, viral clearance is influenced by host immunity and vaginal microbiota[26,27]. Consequently, no significant association was identified between HPV viral load and CIN lesions in the present study. Of course, this discrepancy might be due to insufficient statistical power caused by a small sample sizes of CIN2 (n = 15) and CIN3 (n = 13) samples. In our cross-sectional study, *PAX1* and *JAM3* methylation levels were more prevalent for both CIN2 and CIN3 compared with normal controls. Once epithelial cells are infected with hrHPV, viral DNA can integrate into the host cellular genome, resulting in sustained overexpression of E6 and E7 oncoproteins. These proteins not only inhibit the p53/Rb pathway but also recruit DNA methyltransferases (DNMT1/3B), inducing local hypermethylation at host promoter regions. Consequently, CpG hypermethylation silences the *PAX1* (20p11.2) and JAM3 (11q25) promoters, disabling their regulatory roles in cell differentiation and tight junction-mediated anti-metastasis, respectively, and thereby promoting the progression of cervical lesions[28,29]. Previous studies have confirmed that $PAX1^m$ and $JAM3^m$ exhibited higher positive rates in CC tissues compared with benign cervical tissues, with the methylation levels increasing alongside the severity of CIN lesions. Significantly decreased ΔCt values of both genes were observed from CIN2 to CIN3 lesions. For adenocarcinoma, ΔCt values of *PAX1* (1.78) and *JAM3* (3.4) were markedly decreased[13]. These results indicated that $PAX1^m$ and $JAM3^m$ might be correlated with higher-grade cervical lesions and bad potential as potential adjunctive markers for early CC screening.

Meanwhile, this study also explored the correlations between $PAX1^m/JAM3^m$ and HPV viral load. The results demonstrated that neither $JAM3^m$ nor $PAX1^m$ levels were correlated with all HPV, HPV16/18 or non-HPV16/18 viral load. The present findings were inconsistent with those reported in previous studies[30,31]. The heterogeneity might be explained by different geographical, ethnic and HPV infection characteristics of enrolled participants; inconsistent methylation detection technologies and viral load quantification; and insufficient ability to detect a weak correlation with varied sample sizes. There is also a biological mechanism for HPV viral load, which is largely spatially and temporally heterogeneous and changes with variations in immune status and vaginal microbiome[26,27]. Gene methylation is a relatively stable epigenetic alteration that tends to remain constant in the short term after establishment. The poor temporal adjustment makes it difficult for a cross-sectional study to capture the real correlation between viral load and gene methylation. Additionally, HPV oncogenicity is driven by complex mechanisms, dominated by viral integration and E6/E7 oncoprotein expression. Since HPV viral load only indicates the level of viral replication, it doesn't directly modulate methylation of the host gene. These findings demonstrated that the predictive performance of $PAX1^m/JAM3^m$ detection was not influenced by HPV viral load, which would facilitate the clinical implementation of dual methylation in cervical lesion triage. Regardless of viral load, positive dual gene methylation indicated an increased need for vigorous surveillance, supporting their utility as independent and complementary markers for CC screening. Besides, the $PAX1^m$ had a significant positive association with $JAM3^m$, indicating that the two genes might exert synergistic effects in the occurrence and progression of cervical lesions, cell death, and regulation of tumor invasion and metastasis.

There were some limitations of the present investigation. Firstly, the cross-sectional study design limited causal inference, and the dynamic temporal relationship between HPV viral load during, dual methylation, and cervical lesion progression were not fully elucidated. Secondly, the exclusion of those younger than 30 years of age failed to cover the first infection peak. Thirdly, the limited number of high grade CIN cases weakened the statistical power for analysis. Further prospective studies should validate the relationship between temporal dynamics of HPV viral load and dual methylation.

Conclusions

This research not only supplemented accurate epidemiological evidence for regional CC prevention and control in Changshu but also demonstrated that *PAX1*^{m}*/JAM3*^{m} could represent promising candidate biomarkers for cervical lesions, regardless of HPV viral load.

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Data availability All the data and materials are available from the First author YQX upon request.

Declarations

Competing interests. The authors declare no competing interests.

Ethics approval and consent to participate The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of Changshu Hospital of Traditional Chinese Medicine.

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